

```
centerX = (frameWidth - panelWidth) / 2;
centerY = 270;

panel panel = new JPanel();
panel.setLayout(null);
panel.setBounds(centerX, centerY, panelWidth, panelHeight);
panel.setBackground(new Color(255, 255, 220));
background.add(panel);

// Username
JTextField username = new JTextField();
username.setBounds(40, 40, 240, 40);
panel.add(username);

// Password
JPasswordField password = new JPasswordField();
password.setBounds(40, 100, 240, 40);
panel.add(password);

// Login Button
JButton login = new JButton("Login");
login.setBounds(40, 160, 240, 45);
panel.add(login);

login.setVisible(true);

login.addActionListener(e -> {

    if (username.getText().equals("admin") &&
        String.valueOf(password.getPassword()).equals("1234")) {

        new HomePage();
        dispose();
    }
}
```

```
} else {  
    JOptionPane.showMessageDialog(this, "Invalid  
}
```

```
static void main(String[] args) {  
new LoginFrame();
```

```
ax.swing.*;
a.awt.*;

ss HomePage extends JFrame {

    HomePage() {

        tSize(900, 600);
        tLocationRelativeTo(null);
        tDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        ===== BACKGROUND IMAGE =====
        JLabel background = new JLabel(
            ..... new ImageIcon(getClass().getResource("/images/home.

ckground.setLayout(null);
tContentPane(background);

        ===== TITLE =====
        JLabel title = new JLabel("HOME PAGE", SwingConstants.CENTE
        tle.setFont(new Font("Arial", Font.BOLD, 40));
        tle.setForeground(Color.WHITE);
        tle.setBounds(0, 60, 900, 60);
        ckground.add(title);

        ===== BUTTONS =====
        JButton add = new JButton("Add Record");
        JButton update = new JButton("Update Record");
        JButton delete = new JButton("Delete Record");
        JButton view = new JButton("View Records");
        JButton logout = new JButton("Logout");

        t buttonWidth = 250;
        t buttonHeight = 45;
        t centerX = (900 - buttonWidth) / 2;
```



```
delete.setBounds(centerX, 270, buttonWidth, buttonHeight);  
view.setBounds(centerX, 330, buttonWidth, buttonHeight);  
logout.setBounds(centerX, 390, buttonWidth, buttonHeight);
```

```
background.add(add);  
background.add(update);  
background.add(delete);  
background.add(view);  
background.add(logout);
```

```
setVisible(true);
```

```
// ===== BUTTON ACTIONS =====
```

```
add.addActionListener(e -> new AddRecord());  
update.addActionListener(e -> new UpdateRecord());  
delete.addActionListener(e -> new DeleteRecord());  
view.addActionListener(e -> new ViewRecords());
```

```
logout.addActionListener(e -> {  
    new LoginFrame();  
    dispose();  
});
```

```
}
```

```
import java.sql.*;

public class AddRecord extends JFrame {

    JTextField id, name;
    JComboBox<String> courseBox, deptBox, semBox;
    JButton save;

    public AddRecord() {

        setTitle("Add Student Record");
        setExtendedState(JFrame.MAXIMIZED_BOTH);    // Full screen
        setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

        // ===== GET SCREEN SIZE =====
        Dimension screenSize = Toolkit.getDefaultToolkit().getScreenSize();
        int screenWidth = screenSize.width;
        int screenHeight = screenSize.height;

        // ===== LOAD & SCALE BACKGROUND IMAGE =====
        ImageIcon icon = new ImageIcon(getClass().getResource("/images/add.jpeg"));
        Image img = icon.getImage();
        Image scaledImg = img.getScaledInstance(screenWidth, screenHeight, Image.SCALE_SMOOTH);
        JLabel background = new JLabel(new ImageIcon(scaledImg));
        background.setLayout(null);
        setContentPane(background);

        // ===== TITLE =====
        JLabel title = new JLabel("ADD STUDENT RECORD", SwingConstants.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 50));
        title.setForeground(Color.WHITE);
        title.setBounds(0, 80, screenWidth, 70);
```

```
int panelHeight = 350;

int centerX = (screenWidth - panelWidth) / 2;
int centerY = (screenHeight - panelHeight) / 2;

JPanel panel = new JPanel();
panel.setLayout(null);
panel.setBounds(centerX, centerY, panelWidth, panelHeight);
panel.setBackground(new Color(255, 255, 255, 220));
background.add(panel);

// Labels
JLabel l1 = new JLabel("Student ID");
JLabel l2 = new JLabel("Student Name");
JLabel l3 = new JLabel("Course");
JLabel l4 = new JLabel("Department");
JLabel l5 = new JLabel("Semester");

id = new JTextField();
name = new JTextField();

String[] courses = {"BCA", "BSc", "BCom", "BA", "BBA"};
courseBox = new JComboBox<>(courses);

String[] departments = {"Computer Science", "Commerce", "Mathematics", "Physics", "English"};
deptBox = new JComboBox<>(departments);

String[] semesters = {"Sem 1", "Sem 2", "Sem 3", "Sem 4", "Sem 5"};
semBox = new JComboBox<>(semesters);
```



```
/ Set bounds inside panel
l1.setBounds(40, 30, 150, 30);
l2.setBounds(200, 30, 150, 30);

l2.setBounds(40, 70, 150, 30);
name.setBounds(200, 70, 150, 30);

l3.setBounds(40, 110, 150, 30);
courseBox.setBounds(200, 110, 150, 30);

l4.setBounds(40, 150, 150, 30);
deptBox.setBounds(200, 150, 150, 30);

l5.setBounds(40, 190, 150, 30);
semBox.setBounds(200, 190, 150, 30);

save.setBounds(120, 250, 160, 45);

panel.add(l1); panel.add(id);
panel.add(l2); panel.add(name);
panel.add(l3); panel.add(courseBox);
panel.add(l4); panel.add(deptBox);
panel.add(l5); panel.add(semBox);
panel.add(save);

setVisible(true);

save.addActionListener(e -> addstudent());
```

```
student() {  
  
    con = DatabaseConnection.getConnection();  
  
    sql = "INSERT INTO students_info (student_id, student_name, course, department, semester) VALUES (" +  
        id.getText() + ", " + name.getText() + ", " + courseBox.getSelectedItem().toString() + ", " +  
        deptBox.getSelectedItem().toString() + ", " + semBox.getSelectedItem().toString() + ")";  
  
    PreparedStatement ps = con.prepareStatement(sql);  
  
    ps.setInt(1, Integer.parseInt(id.getText()));  
    ps.setString(2, name.getText());  
    ps.setString(3, courseBox.getSelectedItem().toString());  
    ps.setString(4, deptBox.getSelectedItem().toString());  
    ps.setString(5, semBox.getSelectedItem().toString());  
  
    ps.executeUpdate();  
  
    JOptionPane.showMessageDialog(this, "Record Added Successfully");  
  
    return true;  
}  
  
catch (Exception ex) {  
    JOptionPane.showMessageDialog(this, ex.getMessage());  
}
```



```
import javax.swing.*;
```

```
import java.awt.*;
```

```
import java.sql.*;
```

```
public class UpdateRecord extends JFrame {
```

```
    JTextField id, name;
```

```
    JComboBox<String> courseBox, deptBox, semBox;
```

```
    JButton update;
```

```
    public UpdateRecord() {
```

```
        setTitle("Update Student Record");
```

```
        setExtendedState(JFrame.MAXIMIZED_BOTH);    // Full screen
```

```
        setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);
```

```
        // ===== GET SCREEN SIZE =====
```

```
        Dimension screenSize = Toolkit.getDefaultToolkit().getScreenSize();
```

```
        int screenWidth = screenSize.width;
```

```
        int screenHeight = screenSize.height;
```

```
        // ===== LOAD & SCALE BACKGROUND IMAGE =====
```

```
        ImageIcon icon = new ImageIcon(getClass().getResource("/images/update.jpeg"));
```

```
        Image img = icon.getImage();
```

```
        Image scaledImg = img.getScaledInstance(screenWidth, screenHeight, Image.SCALE_SMOOTH);
```

```
        JLabel background = new JLabel(new ImageIcon(scaledImg));
```

```
        background.setLayout(null);
```

```
        setContentPane(background);
```

```
JLabel title = new JLabel("UPDATE STUDENT RECORD", SwingConstants.CENTER);
title.setFont(new Font("Arial", Font.BOLD, 50));
title.setForeground(Color.WHITE);
title.setBounds(0, 80, screenWidth, 70);
background.add(title);

// ===== CENTER PANEL =====
int panelWidth = 450;
int panelHeight = 350;

int centerX = (screenWidth - panelWidth) / 2;
int centerY = (screenHeight - panelHeight) / 2;

JPanel panel = new JPanel();
panel.setLayout(null);
panel.setBounds(centerX, centerY, panelWidth, panelHeight);
panel.setBackground(new Color(255, 255, 255, 220));
background.add(panel);

// Labels
JLabel l1 = new JLabel("Student ID");
JLabel l2 = new JLabel("Student Name");
JLabel l3 = new JLabel("Course");
JLabel l4 = new JLabel("Department");
JLabel l5 = new JLabel("Semester");

id = new JTextField();
name = new JTextField();

String[] courses = {"BCA", "BSc", "BCom", "BA", "BBA"};
courseBox = new JComboBox<>(courses);

String[] departments = {"Computer Science", "Commerce", "Mathematics", "Physics", "English"};
```

```
deptBox = new JComboBox<>(departments);

String[] semesters = {"Sem 1", "Sem 2", "Sem 3", "Sem 4", "Sem 5"};
semBox = new JComboBox<>(semesters);

update = new JButton("Update");

// Set bounds inside panel
l1.setBounds(60, 40, 150, 30);
id.setBounds(220, 40, 170, 30);

l2.setBounds(60, 80, 150, 30);
name.setBounds(220, 80, 170, 30);

l3.setBounds(60, 120, 150, 30);
courseBox.setBounds(220, 120, 170, 30);

l4.setBounds(60, 160, 150, 30);
deptBox.setBounds(220, 160, 170, 30);

l5.setBounds(60, 200, 150, 30);
semBox.setBounds(220, 200, 170, 30);

update.setBounds(140, 260, 160, 45);

panel.add(l1); panel.add(id);
panel.add(l2); panel.add(name);
panel.add(l3); panel.add(courseBox);
panel.add(l4); panel.add(deptBox);
panel.add(l5); panel.add(semBox);
panel.add(update);
```



```
update.addActionListener(e -> updateStudent());
```

```
private void updateStudent() {
```

```
    try {
```

```
        Connection con = DatabaseConnection.getConnection();
```

```
        String sql = "UPDATE students_info SET student_name=?, course=?, department=?, semester=? WHERE student_
```

```
        PreparedStatement ps = con.prepareStatement(sql);
```

```
        ps.setString(1, name.getText());
```

```
        ps.setString(2, courseBox.getSelectedItem().toString());
```

```
        ps.setString(3, deptBox.getSelectedItem().toString());
```

```
        ps.setString(4, semBox.getSelectedItem().toString());
```

```
        ps.setInt(5, Integer.parseInt(id.getText()));
```

```
        int rows = ps.executeUpdate();
```

```
        if (rows > 0)
```

```
            JOptionPane.showMessageDialog(this, "Record Updated Successfully");
```

```
        else
```

```
            JOptionPane.showMessageDialog(this, "No Record Found");
```

```
        ps.close();
```

```
        con.close();
```

```
    } catch (Exception ex) {
```

```
        JOptionPane.showMessageDialog(this, ex.getMessage());
```

```
    }
```

```
import java.sql.*;
```

```
public class DeleteRecord extends JFrame {
```

```
    JTextField id;
```

```
    JButton delete;
```

```
    public DeleteRecord() {
```

```
        setTitle("Delete Student Record");
```

```
        setExtendedState(JFrame.MAXIMIZED_BOTH);    // Full screen
```

```
        setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);
```

```
        // ===== GET SCREEN SIZE =====
```

```
        Dimension screenSize = Toolkit.getDefaultToolkit().getScreenSize();
```

```
        int screenWidth = screenSize.width;
```

```
        int screenHeight = screenSize.height;
```

```
        // ===== LOAD & SCALE BACKGROUND IMAGE =====
```

```
        ImageIcon icon = new ImageIcon(getClass().getResource("/images/delete.jpeg"));
```

```
        Image img = icon.getImage();
```

```
        Image scaledImg = img.getScaledInstance(screenWidth, screenHeight, Image.SCALE_SMOOTH);
```

```
        JLabel background = new JLabel(new ImageIcon(scaledImg));
```

```
        background.setLayout(null);
```

```
        setContentPane(background);
```

```
        // ===== TITLE =====
```

```
        JLabel title = new JLabel("DELETE STUDENT RECORD", SwingConstants.CENTER);
```

```
        title.setFont(new Font("Arial", Font.BOLD, 50));
```

```
        title.setForeground(Color.WHITE);
```

```
int centerX = (screenWidth - panelWidth) / 2;
int centerY = (screenHeight - panelHeight) / 2;

JPanel panel = new JPanel();
panel.setLayout(null);
panel.setBounds(centerX, centerY, panelWidth, panelHeight);
panel.setBackground(new Color(255, 255, 255, 220));
background.add(panel);

// Label
JLabel l1 = new JLabel("Enter Student ID to Delete");
l1.setBounds(70, 30, 250, 30);

// TextField
id = new JTextField();
id.setBounds(70, 70, 250, 35);

// Button
delete = new JButton("Delete");
delete.setBounds(120, 130, 150, 40);

panel.add(l1);
panel.add(id);
panel.add(delete);

setVisible(true);
```



```
if (id.getText().trim().isEmpty()) {  
    JOptionPane.showMessageDialog(this, "Please Enter Student ID");  
    return;  
}  
  
try {  
  
    Connection con = DatabaseConnection.getConnection();  
  
    if (con == null) {  
        JOptionPane.showMessageDialog(this, "Database Not Connected");  
        return;  
    }  
  
    String sql = "DELETE FROM students_info WHERE student_id=?";  
    PreparedStatement ps = con.prepareStatement(sql);  
  
    ps.setInt(1, Integer.parseInt(id.getText()));  
  
    int rows = ps.executeUpdate();  
  
    if (rows > 0) {  
        JOptionPane.showMessageDialog(this, "Record Deleted Successfully");  
        id.setText("");  
    } else {  
        JOptionPane.showMessageDialog(this, "No Record Found With That ID");  
    }  
}
```

```
s.close();
```

```
on.close();
```

```
catch (NumberFormatException ex) {
```

```
OptionPane.showMessageDialog(this, "ID must be a number");
```

```
catch (Exception ex) {
```

```
ex.printStackTrace();
```

```
OptionPane.showMessageDialog(this, ex.getMessage());
```

```
rt javax.swing.*;
rt javax.swing.table.DefaultTableModel;
rt javax.swing.table.JTableHeader;
rt java.awt.*;
rt java.sql.*;

ic class ViewRecords extends JFrame {

private Image backgroundImage;

public ViewRecords() {

    setTitle("View Records");
    setSize(900, 600);
    setLocationRelativeTo(null);
    setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

    // ===== LOAD IMAGE =====
    backgroundImage = new ImageIcon(
        getClass().getResource("/images/view.jpeg")
    ).getImage();

    // ===== BACKGROUND PANEL =====
    JPanel backgroundPanel = new JPanel() {
        protected void paintComponent(Graphics g) {
            super.paintComponent(g);
            g.drawImage(backgroundImage, 0, 0,
                getWidth(), getHeight(), this);
        }
    };

    backgroundPanel.setLayout(new BorderLayout());
    setContentPane(backgroundPanel);
```



```
===== TABLE MODEL =====
ng[] columnNames = {
    "ID", "Name", "Course", "Department", "Semester", "GPA"
};

JTableModel model =
    new DefaultTableModel(columnNames, 0);

JTable table = new JTable(model);

===== MAKE TABLE TRANSPARENT =====
table.setOpaque(false);
table.setBackground(new Color(0,0,0,0));
table.setForeground(Color.RED); // ● DATA COLOR
table.setFont(new Font("Arial", Font.BOLD, 16));
table.setRowHeight(28);
table.setShowGrid(false);
table.setFillViewportHeight(true);

===== TABLE HEADER STYLE =====
JTableHeader header = table.getTableHeader();
header.setBackground(Color.BLACK);
header.setForeground(Color.RED);
header.setFont(new Font("Arial", Font.BOLD, 18));

JScrollPane scrollPane = new JScrollPane(table);
scrollPane.setOpaque(false);
scrollPane.getViewport().setOpaque(false);

JPanel groundPanel.add(scrollPane, BorderLayout.CENTER);
```

```
// ===== DATABASE =====  
try {  
    Connection con =  
        DatabaseConnection.getConnection();  
  
    Statement st = con.createStatement();  
    ResultSet rs =  
        st.executeQuery("SELECT * FROM students_1");  
  
    while (rs.next()) {  
        model.addRow(new Object[]{  
            rs.getInt("student_id"),  
            rs.getString("student_name"),  
            rs.getString("course"),  
            rs.getString("department"),  
            rs.getString("semester")  
        });  
    }  
  
    rs.close();  
    st.close();  
    con.close();  
  
} catch (Exception ex) {  
    JOptionPane.showMessageDialog(null,  
        "Database Error: " + ex.getMessage());  
}  
  
setVisible(true);
```


COLLEGE RECORD SYSTEM

Login Page

HOME PAGE

Add Record

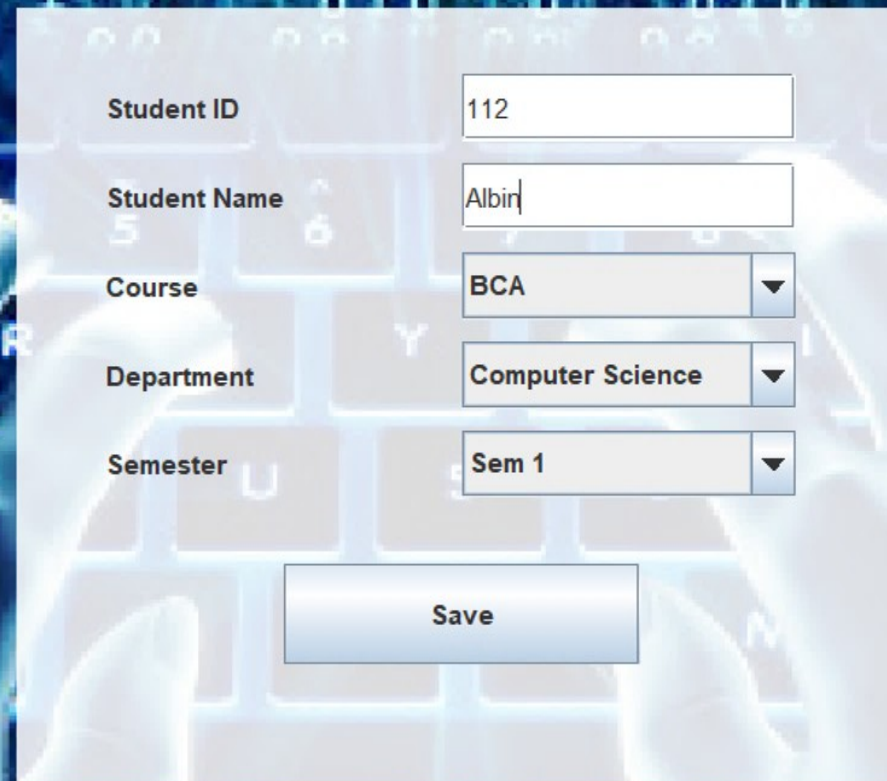
Update Record

Delete Record

View Records

Logout

ADD STUDENT RECORD



Student ID	112
Student Name	Albin
Course	BCA
Department	Computer Science
Semester	Sem 1
Save	

UPDATE STUDENT RECORD

Student ID

Student Name

Course

BCA ▼

Department

Computer Science ▼

Semester

Sem 1 ▼

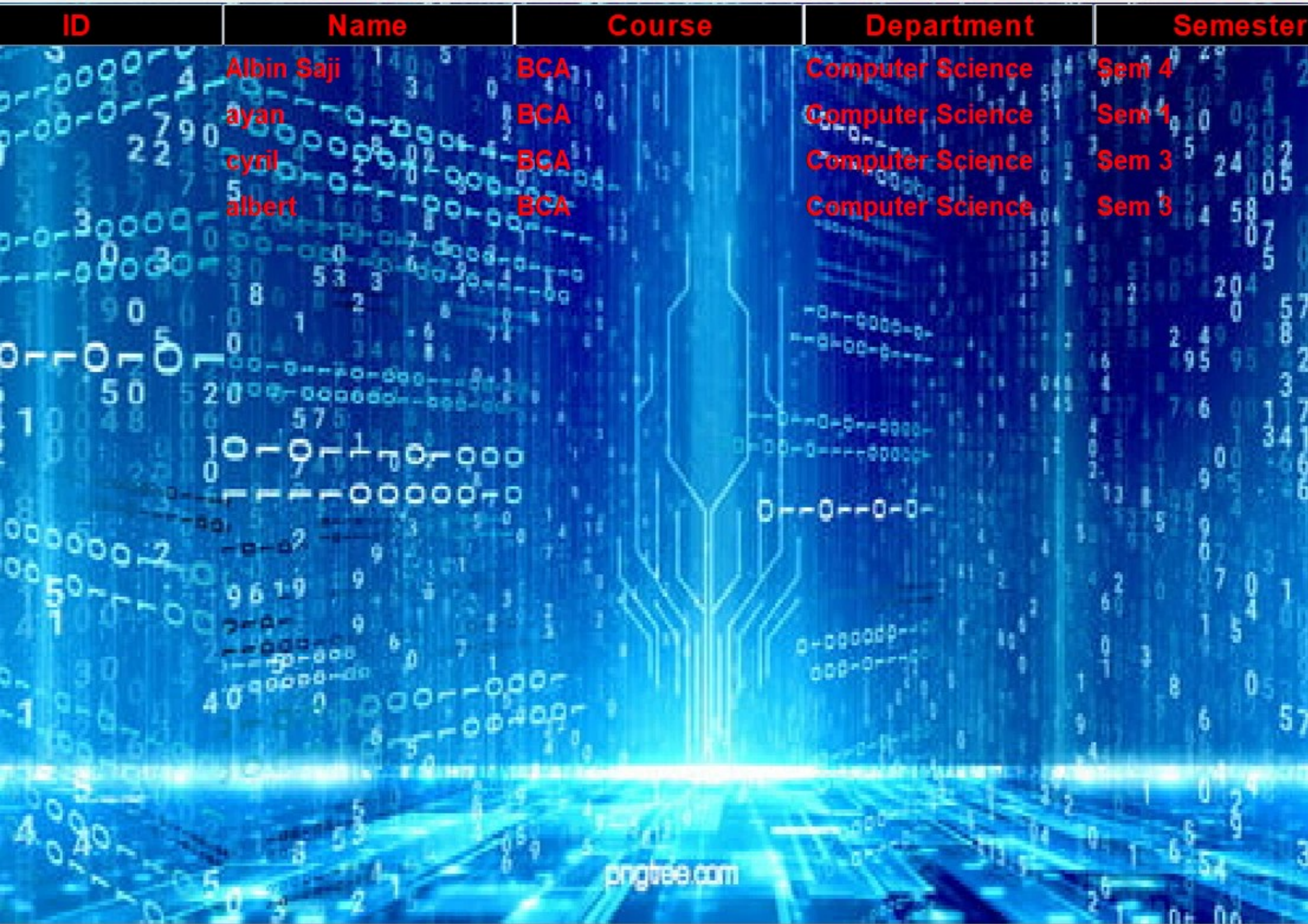
Update

DELETE STUDENT RECORD

Enter Student ID to Delete

Delete

alamy



ID	Name	Course	Department	Semester
	Albin Saji	BCA	Computer Science	Sem 4
	ayan	BCA	Computer Science	Sem 4
	cyril	BCA	Computer Science	Sem 3
	albert	BCA	Computer Science	Sem 3

```
public class DatabaseConnection {  
  
    public static Connection getConnection() {  
  
        try {  
            Class.forName("com.mysql.cj.jdbc.Driver");  
  
            return DriverManager.getConnection(  
                "jdbc:mysql://localhost:3306/college_db",  
                "root",  
                "AACCA@123"  
            );  
  
        } catch (Exception e) {  
            System.out.println(e);  
            return null;  
        }  
    }  
}
```



```
import java.awt.*;
```

```
public class LoginFrame extends JFrame {
```

```
    public LoginFrame() {
```

```
        setSize(1000, 650);    // Good laptop size
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
```

```
        // Background Image
```

```
        JLabel background = new JLabel(new ImageIcon(getClass().getResource("/images/login.jpg")));
        background.setLayout(null);
        setContentPane(background);
```

```
        int frameWidth = 1000;
```

```
        // ===== MAIN TITLE =====
```

```
        JLabel title = new JLabel("COLLEGE RECORD SYSTEM", SwingConstants.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 48));    // MORE BIG
        title.setForeground(Color.WHITE);
        title.setBounds(0, 100, frameWidth, 70);
        background.add(title);
```

```
        // ===== SUB TITLE =====
```

```
        JLabel subtitle = new JLabel("Login Page", SwingConstants.CENTER);
        subtitle.setFont(new Font("Arial", Font.BOLD, 30));    // Bigger
        subtitle.setForeground(Color.WHITE);
        subtitle.setBounds(0, 170, frameWidth, 50);
        background.add(subtitle);
```

```
        // ===== LOGIN PANEL (CENTER CALCULATION) =====
```

```
        int panelWidth = 320;
```